Earth Observation–Driven Monitoring of Regenerative Agriculture:

Satellite–UAV Fusion for Practice Verification and Landscape Biodiversity Indicators

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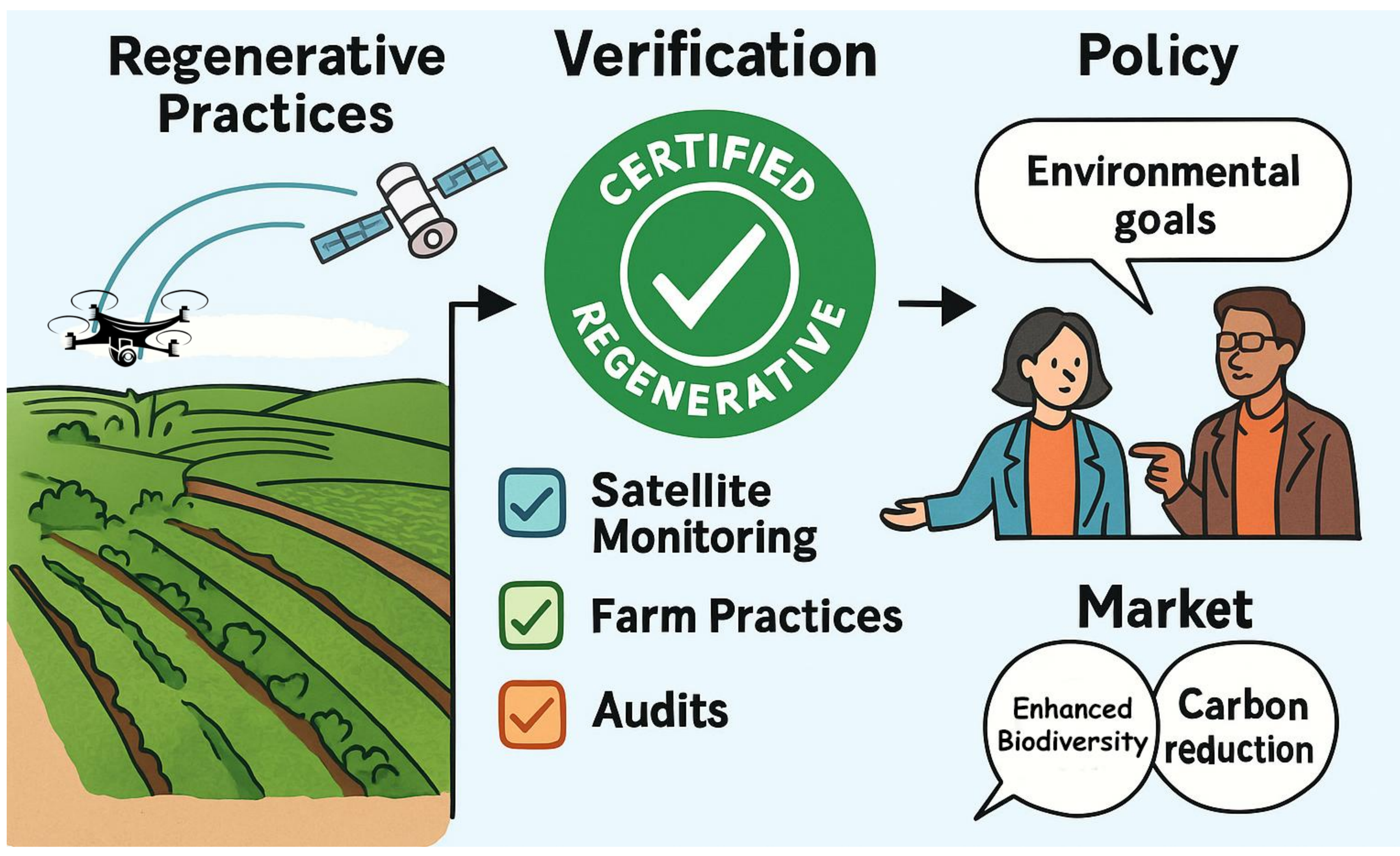
Study Areas

The study spans ~10–15 NaturaSì farms across Italy selected to cover climate/soil gradients, different management practices, hedgerow density, and the presence of ponds/canals/ditches. Italy's agricultural mosaics interlace fields with woody margins and water features, making them ideal to pair satellite-dated practice events with UAV 3D habitat structure.

Targeted SDGs

Aims

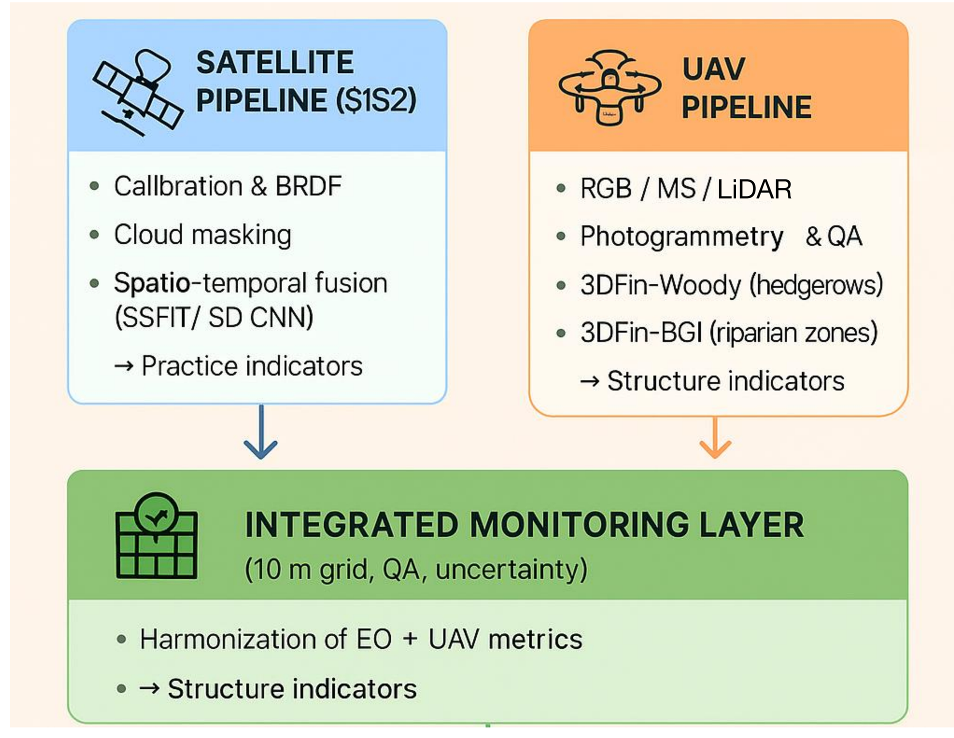
- Build an operational, auditable workflow to verify practices (rotation, cover crops, residue windows, tillage).
- Derive biodiversity-relevant structure along hedgerows & riparian/Blue-Green Infrastructures (BGI) along with continuity, P90 height, bank slope, permanence.
- Package outputs with provenance, QA flags, uncertainty, and admin-level aggregation for policy and certification.



Methods

I
Scope

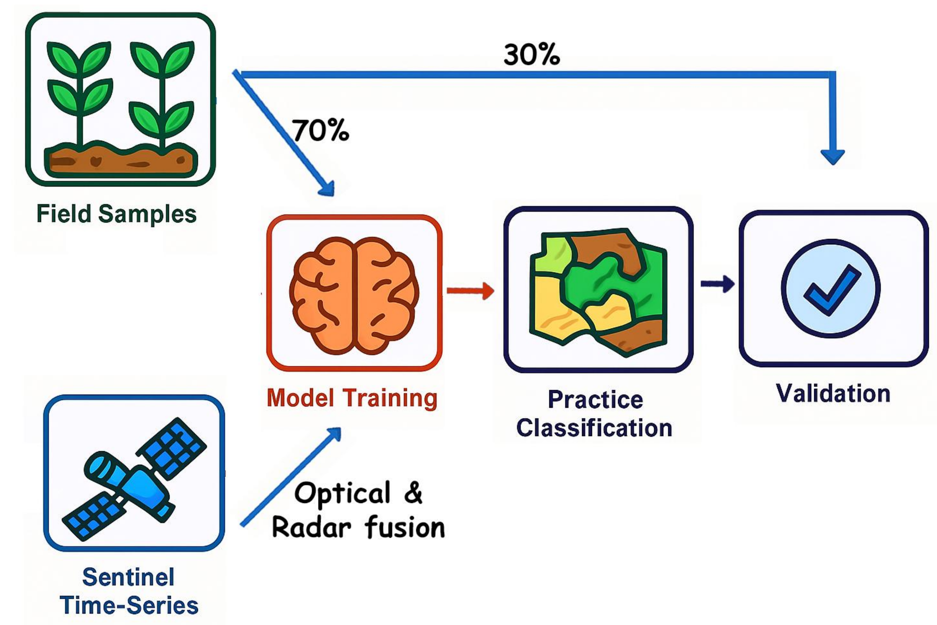
A multiscale Earth Observation workflow integrates Sentinel-2 spectral indices and Sentinel-1 backscatter with UAV 3D to 1) verify field practices and 2) quantify habitat structure along hedgerows and riparian/BGI.



Multiscale EO workflow: Sentinel-1/2 time series + targeted UAV 3D to verify practices and quantify linear-habitat/BGI structure.

II
Satellite time series & fusion

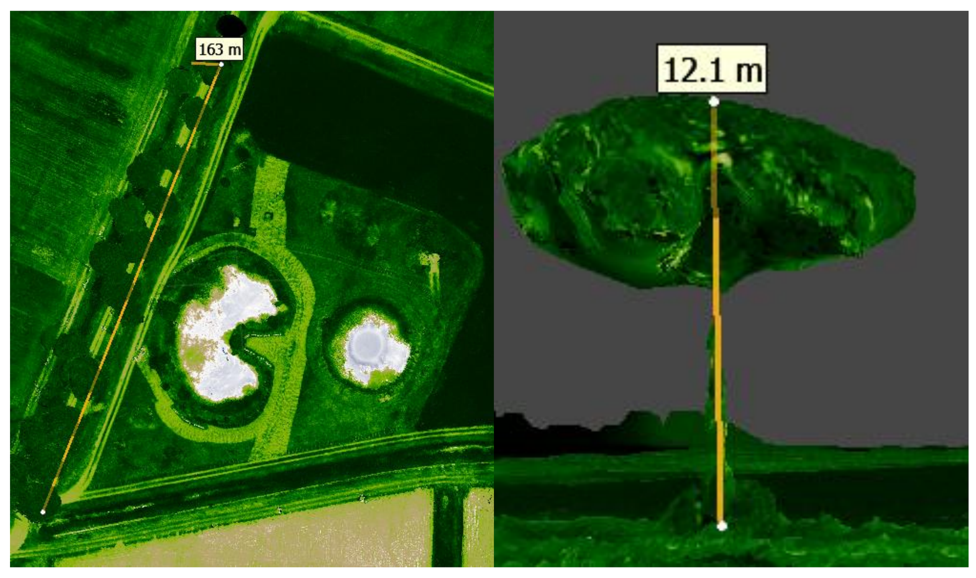
We assemble cloud-robust, calendar-aware SAR–optical time series. A phenology-guided fusion smooths parcel trajectories and flags rapid changes to date cover-crop establishment/termination, compute residue-day totals, and distinguish conservation vs. conventional tillage. Outputs are practice timelines with per-event quality flags.



Phenology-aware SAR–optical fusion models for training and validation of crop rotation, cover-crop, residue, and tillage events.

III
UAV acquisition & 3D products

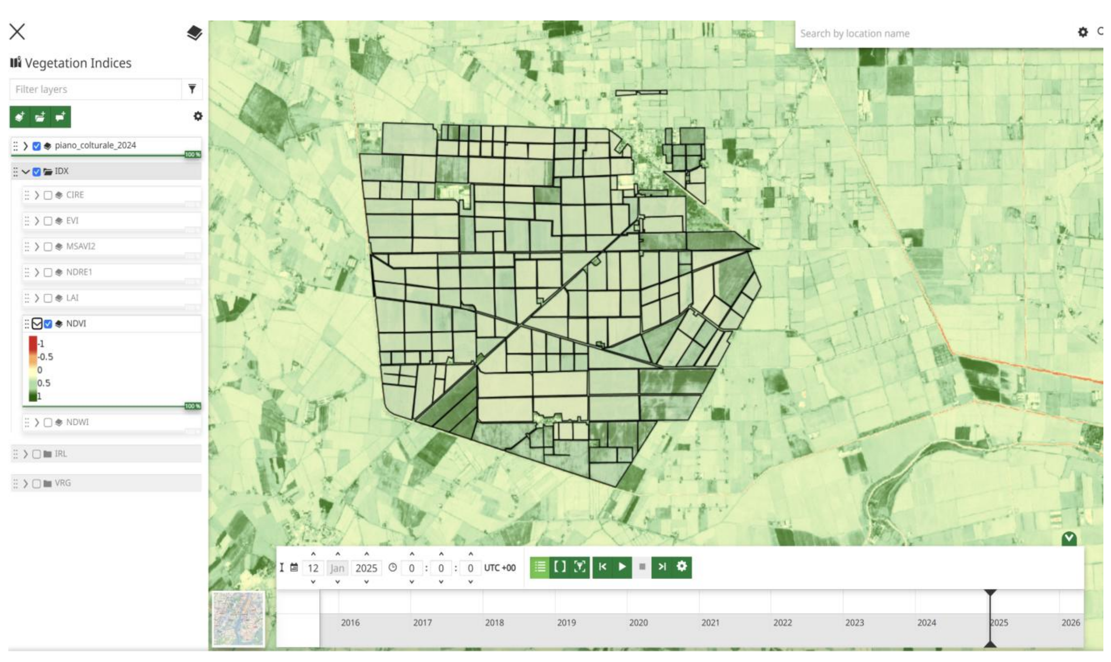
Targeted RGB / MS / LiDAR acquisitions generate digital terrain models / canopy height models (DTM/CHM) and 3D models. We derive hedgerow continuity (1–gap/length), height (mean/P90), overhang, edge/junction density, and for BGI: extent/shape, bank slope/height, woody–water contact, and permanence class.



Examples of geo-features that can be automatically extracted from photogrammetry- and LiDAR-based 3D models, serving as proxies for biodiversity and connectivity.

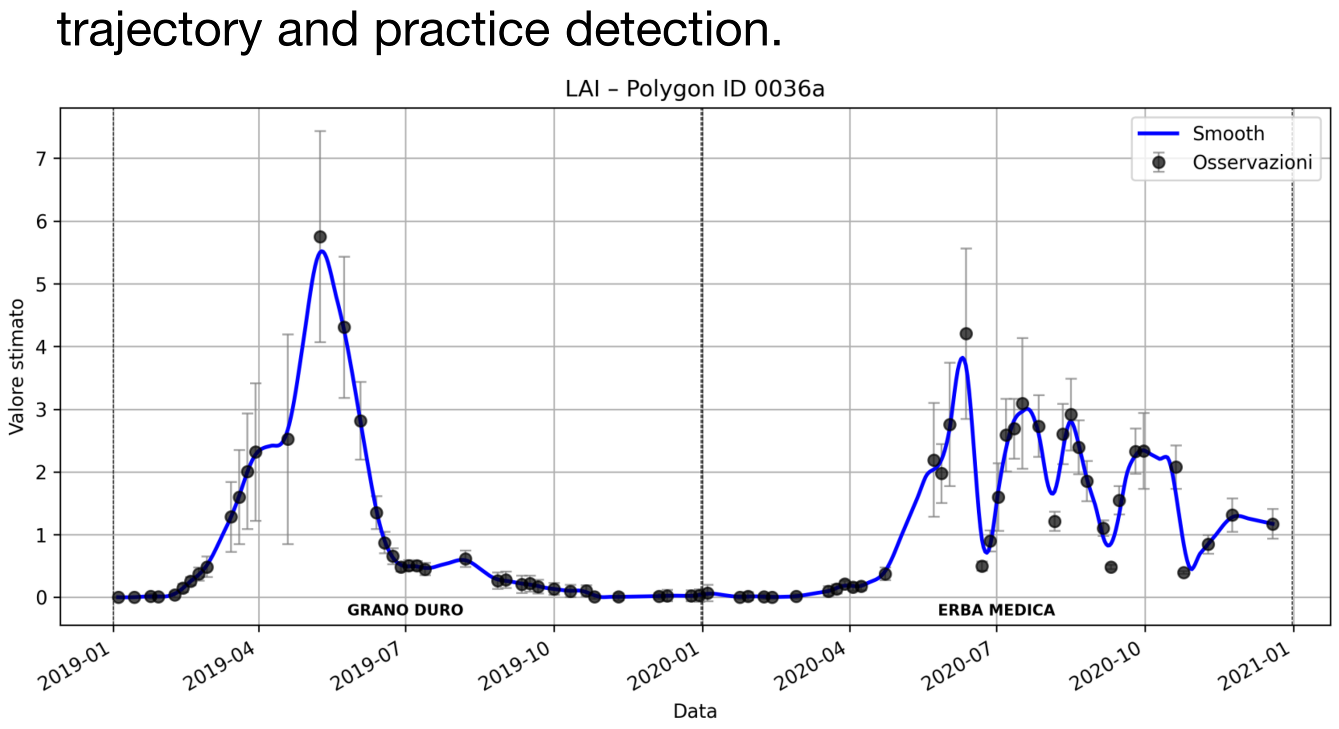
Expected Outcomes

We publish raster time series of vegetation indices and biophysical parameters through a WebGIS built with GeoServer and MapStore for intuitive exploration. Analysts can consume the same layers via Open Geospatial Consortium (OGC) services (e.g., WMS/WFS/WCS) for scripting and reproducible workflows.



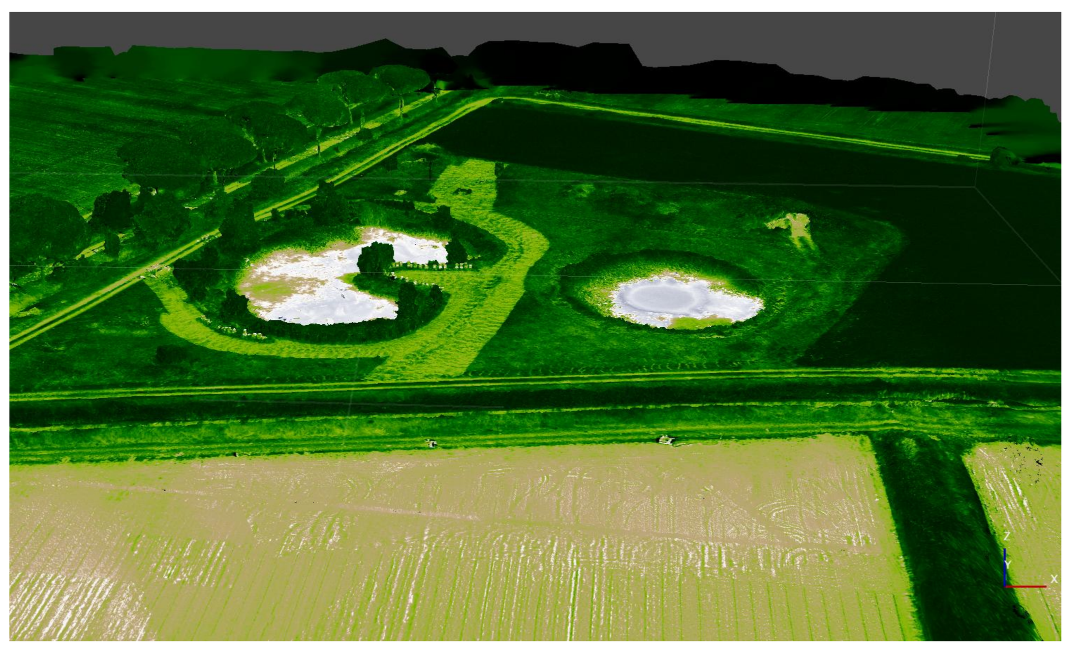
Raster time series of vegetation indices & biophysical parameters served in a user-friendly WebGIS

Leaf Area Index (LAI) time series from Sentinel 2 is used to capture phenology and time-stamp management events. We train models to recognize practice regimes (conventional vs. regenerative). Sentinel-1 backscatter is fused with the optical record to bridge cloudy intervals and enhance sensitivity to soil roughness and moisture, yielding a more reliable, gap-tolerant trajectory and practice detection.



Example using Sentinel-2 LAI time series date cover-crop, residue, and tillage events.

We co-register RGB, LiDAR, and multispectral datasets to produce 3D models, validate them against ground truth data, and develop an integration for 3DFin (Laino et al., 2022). The modules will automatically extract geo-located features from 3D point clouds, DTM and CHM providing proxies of biodiversity and their connectivity across the farmland landscape.



3D model extracted from MSI drone flight with orthogonal and oblique photogrammetry.

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